

GC University Lahore

EXAMINATION: FINAL-TERM 2021

COURSE CODE: CS-4207

TITLE: COMPILER CONSTRUCTION SEMESTER: Spring TIME ALLOWED: 210 Minutes MAX MARKS: 60

Start: 12:30 PM

Paper receiving deadline: 4:00 PM (210 Minutes exact)

Note. Answer any **FIVE** questions. Please note that Open-Book, Open Slides, Online exam Provision is for saving student's time. It does not mean that answers can be copied or shared. Books and Online resources are for help and hints, not for solutions. Even if an exact answer of any question is present in a book or internet, copy-pasting an exact copy of any book or internet resource will result in a UMC. Mutual solving in groups, phone or online collaboration is dishonesty.

Exam will be handwritten or typed and photographed/scanned and converted to single pdf file. All pages must be numbered. Kindly make black and white photo/pdf file to save space and time. Kindly thoroughly check each page of final converted pdf before sending. Kindly CC the email to yourself also. File can be accepted via email only. WhatsApp or Team uploading/sending is not accepted.

Send it by email attachment to your relevant class teacher (ifakhir@yahoo.com). File name must be your roll number in xxx-BSCS-2018.pdf format. Also mention your roll number in email subject. No other means of submission is accepted.

Q.1: a) From the given CFG find FIRST, FOLLOW sets of each non-terminals and draw LL(1) parsing table: (8)

$S \rightarrow UVW$

$U \rightarrow (S) \mid aSb \mid d$

$V \rightarrow aV \mid \epsilon$

$W \rightarrow cW \mid \epsilon$

b) Draw DAG with construction steps using SDD for the expression: (4)

$x = a + a * (c + d) + (c + d) * a - c / d$

Q.2: a) Explain in detail the process of compilation. Illustrate the output of each phase of compilation for the input: (6)

$x = (a - c) * (c + d) / 2$

b) Describe the Token, Pattern & Lexeme with example. Write pattern for signed number. (6)

Q.3: a) How many types of error can identify and locate by a good compiler? (5)

b) Draw a top-down parse tree for the input **abaccdcd** with complete steps by using following grammar. (7)

$S \rightarrow aAd \mid aB$

$A \rightarrow bS \mid cB$

$B \rightarrow ccdA \mid d$

Q.4: a) Write Postfix notation and draw Abstract Syntax Tree for the input: (6)

$x = a * -c + c * d - e * f + e$

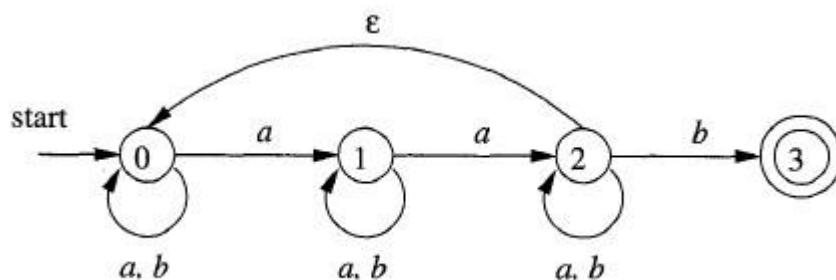
b) Generate Syntax Directed Definition semantics of the given Grammar and draw annotated parse tree for $4 - 7 / 9$: (6)

$S \rightarrow TS'$

$S' \rightarrow -TS' \mid \varepsilon$
 $T \rightarrow MT'$
 $T' \rightarrow /MT' \mid \varepsilon$
 $M \rightarrow num$

OR

b) Simulate given NFA on input aabb



Q.5: a) Tokenize the following program:

(4)

```

int prod (int a[], int b[])
{ int d = 0, i;
for(i = 0; i < 20; i++)
{ d+= a[i] * b[i]; }
return d; }

```

b) Explain cross compilation process in detail. Why and when it is necessary.

(4)

c) Differentiate front-end and back-end of a compiler. In which situations front-end and back-end may change.

(4)

Q.6: a) Draw DFA and LR(1) parsing table for the following grammar:

(8)

$S \rightarrow L = R \mid R$
 $L \rightarrow -R \mid id$
 $R \rightarrow L$

b) Draw working and parse tree for the input $id = -id$, by using LR(1) parsing table of part a).

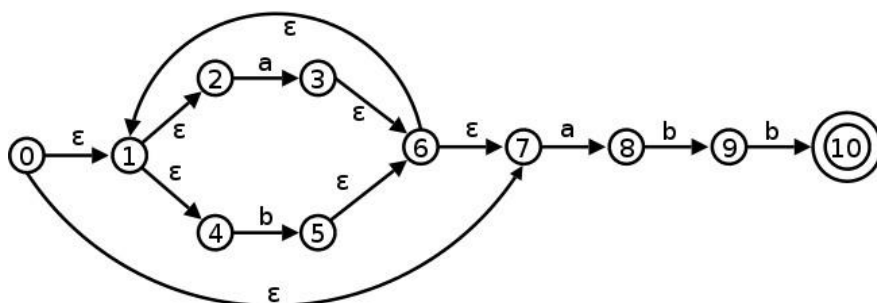
(4)

OR

b) Explain how to convert an NFA to a DFA using subset construction algorithm.

Q. 7: a) Generate Regular expression from given NFA. Final answer is not important, but steps will be marked.

(6)



b) Draw NFA which represents given regular expression $(a|b)^*ac$. Final answer is not important but steps will be marked.

**Department of Computer Science
Final Examination Review Report**

Compiler Construction
Start Time : 14-07-2020 @ 11am

Semester VI
Deadline : 14-07-2020 @ 05:00 PM

Write your answers in report form to address following topics/questions?

1. At what stage of compilation panic mode and phrase level recovery is activated.
2. Select a suitable example and discuss all the phases of compiler construction
3. Use any CFG to explain how FIRST and FOLLOWS are computed.
4. Use your example and construct a parse table for LL(1) Parser using the grammar (part 3) and First and Follows computed for the same grammar. Use any suitable example to verify using your parse table
5. What is the significance of lexeme in compiler construction? Use following example and divide the code into appropriate lexemes.

```
float getVal(int x){  
    return (x>10 ? 10 : x/2);  
}
```
6. What is backtracking why it is not considered very fruitful?
7. Differentiate Top Down parsing and Bottom Up Parsing
8. In Bottom Up Parsing why shift reduce is considered more general than other parsing.
9. Illustrate the shift-reduce and reduce-reduce conflict with a suitable example.
10. Use the following example and verify that given grammar is LR(0), SLR(1), LALR(1) or CLR(1). You need to show all steps in detail to support your answer.

$E \rightarrow E + T \mid T$

$T \rightarrow TF \mid F$

$F \rightarrow F^* \mid a \mid b$

Show parse table of each of the Parser.

NOTE: Max file size is 5MB. Use Microsoft word. If you want to submit handwritten report then make its pdf and then submit it. Only Microsoft word and pdf format is acceptable.

Best of Luck

GC University Lahore

BS Computer Science Examination: Final 2022

Semester: 5th

COURSE CODE: CS-4207

TITLE: Compiler Construction

Time allowed: 155 minutes

Marks: 40

Subjective

Attempt any Four Questions

Please provide Correct Question No and Part while attempting on answer sheet. Provide only precise answer as asked. Avoid unnecessary details. ---

Question No. 02:

A. Construct unambiguous grammar for Right Associative lists of identifier separated by commas. (01)

B. Construct a syntax-directed translation scheme that translates arithmetic expressions from postfix notation into infix notation. Give annotated parse trees for the input $95 - 2*$. (06)

C. Do Left Factoring in the following: (03)

C. Do Left Factoring in the following: (03)

$\text{Stmt} \rightarrow \text{if Exp then Stmt} \mid \text{if Exp then Stmt else Stmt} \mid \text{a}$

$\text{Exp} \rightarrow \text{b}$

Question No. 03:

A. Compute FIRST and FOLLOW for the following CFG: (03)

$S \rightarrow Aa \mid Bab$

$A \rightarrow BB \mid c \mid \varepsilon$

$B \rightarrow Sd \mid e \mid \varepsilon$

B. For the production, $A \rightarrow XYZ$ compute all LR(0) items. (02)

C. If $A \rightarrow \alpha \cdot B\beta$ is in $\text{CLOSURE}(I)$ and $B \rightarrow \gamma$ is a production then what would be $\text{CLOSURE}(I)$. (02)

D. Consider the following regular expression: (03)

$(a \mid b)^*a(a \mid b)(a \mid b)$

- i. Describe the language denoted by the given regular expression
- ii. Provide transition diagram to recognize the same language of the given regular expression

Question No. 04

Consider the following grammar:

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow (E)$$

$$F \rightarrow \text{id}$$

- A. Compute set of LR (0) item for an augmented expression grammar. (3)
- B. Construct SLR (1) Parse Table (3)
- C. Show the working of Shift-reduce parser to parse input string $(\text{id}+\text{id}) * \text{id}$ (4)

Question No. 05:

Consider the grammar

$$S \rightarrow (L)$$

$$S \rightarrow \text{id} \quad \text{where id is any sequence of alphabets a-z}$$

$$L \rightarrow S$$

$$L \rightarrow L, S$$

- A. Use this grammar to parse through LL(1) Parser (4)
- B. Give right most derivation for the string $((a, a), a, (a))$. (3)
- C. Compute the CLOSURE and GOTO for the given grammar. (3)

Question No. 06:

A. Consider the following grammar

$$A \rightarrow Ba \mid Aa \mid c$$

$$B \rightarrow Bb \mid Ab \mid d$$

Eliminate Left Recursion from the grammar (3)

B. Consider the following grammar

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \varepsilon$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' \mid \varepsilon$$

$$F \rightarrow (E) \mid \text{id}$$

Check whether the grammar is LR(0) or not (5)

GC UNIVERSITY LAHORE

SUBJECTIVE QUESTION PAPER

EXAMINATION: Final 2024

SEMESTER: V

COURSE CODE: CS-4207

SUBJECT: Compiler Construction

TITLE: Compiler Construction

TIME ALLOWED: 135 MINUTES

MAX MARKS: 40

Note: Attempt any FOUR questions from the following. All Questions carry equal marks. Please provide Correct Question No and Part while attempting on answer sheet. Provide only precise answer as asked. Avoid unnecessary details. ---

Question No. 02:

A. Consider the following Grammar and generate DFA that recognizes identifier/Integers. L represents a single letter and D represents a single digit. $_$ is representing underscore. (04)

$$S \rightarrow L \text{ Id} \mid L$$
$$S \rightarrow D \text{ Int} \mid D$$
$$\text{Int} \rightarrow D \text{ Int} \mid D$$
$$\text{Id} \rightarrow D \text{ Id} \mid L \text{ Id} \mid _U$$
$$\text{Id} \rightarrow D \mid L$$
$$U \rightarrow D \text{ Id} \mid L \text{ Id} \mid D \mid L$$

B. Do Left Factoring in the following: (03)

$$\text{Stmt} \rightarrow \text{if Exp then Stmt} \mid \text{if Exp then Stmt else Stmt} \mid a$$
$$\text{Exp} \rightarrow b$$

C. Consider the Following Grammar and eliminate the left Recursion: (03)

$$X \rightarrow XSb \mid Sa \mid b$$
$$S \rightarrow Sb \mid Xa \mid a$$

Question No. 03:

A. Compute FIRST and FOLLOW for the following CFG: **(03)** *(Handwritten note near the text: {a, b} or similar marking)*

$$S \rightarrow Aa \mid Bab$$

$$A \rightarrow BB \mid c \mid \varepsilon$$

$$B \rightarrow Sd \mid e \mid \varepsilon$$

B. For the production, $A \rightarrow XYZ$ compute all LR(0) items. **(02)** C. If $A \rightarrow \alpha \cdot B\beta$ is in $CLOSURE(I)$ and $B \rightarrow \gamma$ is a production then what would be $CLOSURE(I)$. **(02)** D. Construct a DAG and generate quadriuples, triples, and indirect triples for the expression: **(03)**

$$x = b \times -c + b \times -c$$

(Note: There is a hand-drawn Syntax/DAG tree diagram at the bottom of the first page showing operations with nodes labeled with +, ×, −, b, and c, along with some handwritten marks like $\frac{1}{2}$ and "PTC".)

Question No. 04

Consider the following grammar:

(Handwritten notes next to the rules show an explicit right-to-left reduction trace using E, T, F, like \$E, \$T × F, \$F × F, etc.)

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow (E)$$

$$F \rightarrow \text{id}$$

A. Compute set of LR(0) item for an augmented expression grammar. **(03)** B. Construct SLR(1) Parse Table. **(03)** C. Show the working of Shift-reduce parser to parse input string $(\text{id} + \text{id}) * \text{id}$. **(04)** ---

Question No. 05:

Consider the grammar:

$$S \rightarrow (L)$$

$$S \rightarrow \text{id} \quad \text{where id is any sequence of alphabets a-z}$$

$$L \rightarrow S$$

$$L \rightarrow L, S$$

A. Give right most derivation for the string $((a, a), a, (a))$. **(03)** B. Compute the CLOSURE and GOTO for the given grammar. **(03)** C. Draw DAG and show steps of construction using SDD (Syntax Directed Definition) of: **(04)**

$$x = a * a + 2 * a * b + b * b$$

Question No. 06:

A. What are the phases of Compiler Construction? Briefly discuss each section consider the expression: **(04)**

$$\text{position} = \text{initial} + \text{rate} * 60$$

B. Show that the Given grammar is CLR(1) or not. **(06)**

$$S \rightarrow aAd \mid bBd \mid bAe$$

$$A \rightarrow c$$

$$B \rightarrow e$$